Precision-AMR: Spatially Adaptive Floating-Point Precision for PDE-Based Reverse Time Migration

A new axis of numerical adaptivity for adjoint wave-equation solvers, developed on dedicated NVIDIA hardware and validated across heterogeneous NVIDIA and AMD GPU architectures on the Santos Dumont supercomputer (LNCC)

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1 Problem Statement

GPU hardware is increasingly shifting toward low precision in favor of AI workloads (FP8/BF16), exposing new memory hierarchies, tensor cores, and mixed-precision math libraries. At the same time, scientific simulations based on partial differential equations (PDEs) continue to require FP64/FP32 precision to remain trustworthy. Reverse Time Migration (RTM), a seismic imaging method that solves the wave equation by finite differences over thousands of time steps, is a canonical example of this tension: geophysicists use the migrated image to guide expensive drilling decisions, so numerical fidelity is essential. Closely related, Full Waveform Inversion (FWI) estimates the subsurface velocity model itself, iteratively minimizing the difference between observed and simulated seismograms. Its gradient is obtained by the adjoint method, correlating the forward wavefield with the back-propagated adjoint field. RTM and FWI therefore share the same computational core, forward propagation followed by backward propagation with correlation between the two fields, and consequently the same bottleneck: both are memory-intensive, require hundreds of gigabytes of wavefield state, and rely on checkpointing to trade memory against recomputation in that adjoint structure. In FWI the cost is further amplified, because this structure repeats at every inversion iteration.

Our prior work tackled this structure's main bottleneck, namely host–GPU checkpoint transfer over PCIe, by combining the concept of prefetching with data compression. This idea was materialized as the GPUZIP library [16, 18], which compresses checkpoints on the GPU and prefetches them proactively, reducing host–GPU transfer volume by up to $80\times$ and achieving end-to-end speedups of up to $5.1\times$ on V100, while preserving image quality (SSIM ≈ 1, near-zero relative error). Separately, reduced-precision arithmetic has already been shown to be viable for seismic modeling, RTM, and FWI: Fabien-Ouellet [11] found that a stable half-precision (FP16) implementation of the elastic wave equation is achievable through appropriate scaling, that the resulting seismogram error is a negligible fraction of total seismic energy and largely incoherent with seismic phases, and that this noise does not adversely affect FWI or RTM image quality. However, that reduction is applied uniformly across the entire computational domain. Gao and Harms [12] identified a specific hazard of that approach: naive half-precision time stepping accumulates round-off error through a disguised recursive summation in the update rule, degrading solution quality, with compensated (Kahan) summation as an effective remedy.

At the same time, adaptive mesh refinement (AMR) has been explored for seismic wave simulation and, more recently, explicitly for FWI [23, 24] and RTM [7, 15, 17, 29] workflows, to avoid the waste inherent in a uniform grid: dispersion and dissipation errors require grid spacing proportional to the shortest wavelength, so a uniform mesh is forced to use the finest spacing everywhere, even where the local velocity field would tolerate a much coarser one. AMR addresses this by refining Δx locally. Over four decades, since the seminal work of Berger and Colella [5], this idea has consolidated into a well-established taxonomy of adaptivity axes: cell-size refinement (h), polynomial-order enrichment (p), the combination of the two (hp), and vertex relocation without changing the topology (r). All of these axes, however, act on the geometric discretization of the domain [28]. No comparable, physically motivated scheme exists for floating-point precision: existing mixed-precision work in seismic imaging is either global and uniform (as above), or adaptive only at the level of an individual linear-algebra operation (mixed-precision GEMM, iterative refinement, Ozaki-scheme emulation [19]), not tied to a spatial decomposition of the wavefield itself. None of this prior work varies floating-point precision according to physically motivated spatial criteria. A literature survey we conducted on AMR and on AMR applied to RTM/FWI confirms this picture: the works retrieved adapt mesh resolution, order, or position, and none adapts arithmetic precision. This is therefore an axis of adaptivity orthogonal to the established ones, and one that remains largely unexplored.

This project proposes to close that gap with Precision-AMR (P-AMR): a spatially block-structured scheme, an adaptive-precision analogue of AMR, that keeps mesh resolution fixed and instead varies the floating-point format assigned to each block (tile) of the domain, based on physically motivated error-tolerance criteria (proximity to absorbing boundaries, wavefield energy density, adjoint-sensitivity magnitude, and a time-dependent band trailing the propagating wavefront) that predict whether the round-off error introduced in that block can reach the imaging condition or the FWI gradient. This is a structurally different problem from mesh-based AMR. In AMR, the characteristic artifact is a localized reflection at the coarse-fine grid interface, because refinement changes the discretization itself. In P-AMR, no discretization boundary is introduced; instead, a spatially varying level of round-off noise is injected into a hyperbolic PDE, and that noise propagates together with the wave. Tagging criteria therefore cannot rely solely on local smoothness, as AMR refinement indicators do; they must instead approximate where an error, once injected, will decay, be absorbed, or otherwise fail to reach the receivers or the sensitivity kernel before the simulation ends, and control round-off accumulation over the thousands of time steps typical of RTM/FWI.

Because RTM/FWI already require a block/tile decomposition of the domain for checkpointing, P-AMR reuses that same decomposition as its tagging granularity, unifying the precision budget and the lossy-compression budget of differential checkpointing into a single, jointly tuned per-tile accuracy target. Mixed-precision GEMM, cuSOLVER iterative refinement, and Ozaki-scheme FP64 emulation remain essential as supporting techniques, but are now applied selectively, only where P-AMR's criteria indicate they are needed. This is the natural way to exploit GPUs geared primarily toward AI, such as the RTX 5090, which offer negligible native FP64 throughput.

Because the tagging criteria are motivated by the physics of wave propagation and adjoint sensitivity, rather than by any particular vendor's arithmetic units, we treat their portability across architectures as an explicit question, not an assumption. Our research group has access to the Santos Dumont supercomputer (LNCC/SINAPAD), which includes AMD Instinct MI300A accelerators alongside several NVIDIA generations (Volta V100, Hopper H100, Grace-Hopper GH200). This lets us test that question directly, rather than merely assert it, and it is what gives Precision-AMR its potential contribution to the field: a numerical adaptivity axis that is new and portable across vendors, not merely a one-off optimization for a single GPU architecture.

2 Expected Results

The project's principal contribution is the Precision-AMR (P-AMR) method itself, which explores a new axis of numerical adaptivity that varies floating-point precision spatially, according to physically motivated criteria (proximity to the PML, wavefield energy density, adjoint-sensitivity magnitude, and a wavefront-trailing band), proposed and validated specifically for adjoint PDE solvers such as RTM/FWI. This contribution includes the formulation of the four tagging criteria, the characterization of the "precision-interface" artifact, analogous to the coarse-fine interface reflections of classical AMR, and the demonstration that the precision and lossy-compression budgets can be unified into a single per-tile error target. It amounts to adding an arithmetic axis to a taxonomy of adaptivity that, since Berger and Colella [5], has remained entirely geometric [28].

As secondary contributions, the project will produce: (i) a multi-architecture evaluation of P-AMR-guided mixed-precision usage, comparing NVIDIA GPUs (V100, H100, GH200) and the AMD Instinct MI300A APU, characterizing how much of the achieved speedup and accuracy trade-off is vendor-independent; and (ii) the implementation of P-AMR as an **independent library**, with an interface that allows its integration into other PDE-based simulation applications, beyond the specific scope of RTM/FWI. This library may be adopted standalone or combined with GPUZIP, merging P-AMR's spatial precision zoning with GPUZIP's compressed checkpointing into a single workflow, without requiring the host application to adopt the entire GPUZIP stack just to benefit from P-AMR. It will be made available as open source, under the MIT license, on GitHub.

The central scientific result will be submitted to a high-impact high-performance-computing event or journal, for example, Supercomputing (SC), IPDPS, Euro-Par, or SBAC-PAD among the events, or IJHPCA or a Transactions-type journal from IEEE or ACM among the journals, accompanied by a public reproducibility package (datasets, Nsight profiles, quality metrics), made available via UNICAMP's REDU.

3 Scientific and Technological Challenges and the Means and Methods to Overcome Them

The feasibility of adaptive mesh refinement (AMR) for reducing the computational cost of seismic simulation is already well established in the literature. van Driel et al. [26] and Thrastarson et al. [24] show, respectively, that wavefield-adapted meshes accelerate forward/adjoint modelling and FWI by up to an order of magnitude when medium properties vary smoothly. Yang et al. [29] apply a variable-spacing mesh specifically to elastic RTM, preserving multicomponent imaging accuracy while reducing computational cost and storage. Davis and LeVeque [8] further show that refinement indicators guided by the adjoint computation, and not just by local smoothness of the field, produce more efficient mesh refinements in hyperbolic wave-propagation problems, the same logic that motivates P-AMR's sensitivity (FWI) criterion.

This picture is consistent with the broader state of the art in AMR. The review by Vivarelli et al. [28] identifies the refinement indicator as the hardest open question in the field: indicators based on local features of the field are cheap, but tend to over-refine and do not guarantee a reduction of the error in the quantity of interest, whereas adjoint-guided indicators achieve higher accuracy in fewer adaptation steps, at the cost of greater complexity. On the infrastructure side, block-structured decomposition is the granularity that sustains AMR codes at exascale [30], and the interaction between adaptivity and increasing hardware heterogeneity is recognized by the AMR community itself as an additional, still unresolved axis of complexity [10].

None of these works, however, varies *floating-point precision* instead of *mesh resolution*. Nor does any of them characterize empirically how the numerical behavior of an RTM/FWI simulation, that is, the dynamic range of floating-point values actually exercised, distributes across space and time. This double gap, the absence of a physically motivated precision-zoning scheme and the absence of the empirical characterization needed to ground it before designing the criteria, defines this project's scientific and technological challenges, organized in the six phases below, over a 36-month programme.

3.1 Phase 1 — Infrastructure and Baseline (M1–M4)

Challenge: before any precision change is attempted, it is necessary to faithfully reproduce the GPUZIP/Awave-3D baselines under the HPC SDK and ensure that the block decomposition used for checkpointing can also carry a per-block precision tag, an engineering prerequisite that every subsequent phase depends on.

Method: port Awave-3D and GPUZIP to the HPC SDK (nvc++), re-integrate nvCOMP/Bitcomp, and reproduce the V100 baselines on Santos Dumont's V100 partition, matching the original GPUZIP hardware.

- Establish a block/tile decomposition of the domain, extending the existing checkpoint-block structure so that it can also carry a precision tag per block.
- Profile every kernel with Nsight Systems and Nsight Compute, on both the RTX 5090 and Santos Dumont, confirming that FDM stencils are memory-bandwidth-bound at the tile granularity before any precision change is attempted.

3.2 Phase 2 — Empirical Characterization of Mixed-Precision Behavior (M4–M10)

Challenge: the four tagging criteria proposed for Phase 3 are physically motivated, but, unlike the adjoint-guided mesh refinement of Davis and LeVeque [8], no work in the literature empirically characterizes how the dynamic range of floating-point values (exponents and magnitudes actually used) behaves spatiotemporally in an RTM/FWI simulation, nor investigates whether that behavior can be anticipated directly from the input velocity model, before the wave even propagates. Without that characterization, the tagging criteria remain plausible hypotheses that have not yet been verified against actual numerical behavior.

There is, however, an encouraging precedent on the neighboring geometric axis: in seismic AMR, the input velocity model is already used successfully as an *a priori* predictive signal for adapting resolution. Takekawa and Mikada [23] subdivide calculation points located in low-velocity zones; Liu and Zhang [15] use coarse grids in high-velocity zones and fine grids in low-velocity zones for LSRTM; Roberts et al. [21] size elements by the local P-wavespeed and the peak source frequency; Aghamiry et al. [1] tune stencil weights to the local number of grid points per wavelength; and Santos et al. [9] adapt the mesh directly to the velocity field through optimal transport, delineating salt bodies and sharp interfaces without incurring the I/O cost of re-meshing. None of these works, however, tests whether the same signal predicts *tolerance to round-off error*, and not merely the required spatial resolution. It is exactly this transposition, from an already validated geometric axis to the arithmetic axis, that Phase 2 investigates.

Method: instrument the solver to record, from FP64 reference runs, dynamic-range statistics of the wavefield per block and per time step: exponent histogram, minimum/maximum magnitude, variance, across the six datasets described in the Evaluation section.

- Correlate these statistics with attributes derived from the input velocity model (local impedance contrast, depth, distance to reflectors, local wavelength), seeking which attributes are predictive of where the dynamic range, and therefore the tolerance to round-off error, is structurally lower.
- From that correlation, build an *a priori* precision-zoning estimator, computable directly from the velocity model before any time step is run. This estimator is complementary to the four reactive criteria of Phase 3, which depend on the wavefield computed during the simulation itself.
- Use this characterization to validate, or refute, the physical assumptions behind each of the four proposed criteria, before investing in their implementation and fine-tuning in Phase 3.

3.3 Phase 3 — Precision-AMR: Criteria, Tagging, and Validation (M10–M20)

Challenge: unlike mesh-based AMR, where the characteristic artifact is a localized reflection at the coarse-fine interface, round-off noise induced by precision reduction propagates together with the wave itself, so tagging criteria cannot rely solely on local smoothness of the field, as classical refinement indicators do. They must anticipate where an error, once injected, will fail to reach the receivers or the FWI gradient before the simulation ends. A second, technological challenge compounds this: reduced precision and lossy compression, both already used separately in GPUZIP, may interact non-trivially when applied to the same block, and that interaction needs to be isolated before defining a joint error budget.

Method: this is the core scientific phase, run at scale on Santos Dumont's H100 partition. Building on the characterization from Phase 2, we design, implement, and validate four candidate tagging criteria for spatial precision zoning:

- **Boundary criterion:** tiles inside or adjacent to the PML/absorbing layer are tagged for reduced precision by default, since their contribution is already being damped.
- **Energy criterion:** tiles are tagged by a threshold on local wavefield energy density, so that regions of low amplitude (before wavefront arrival, or after geometrical spreading has attenuated the signal) tolerate lower precision.
- **Sensitivity criterion (FWI):** tiles are tagged using the magnitude of the local adjoint-sensitivity kernel, reusing the gradient computation itself as the tagging signal, in direct analogy to the adjoint-guided refinement of Davis and LeVeque [8] and to the truncation-error estimators of classical AMR [5].
- **Wavefront-trailing criterion:** a time-dependent band immediately behind the propagating wavefront is tagged for reduced precision, while the leading edge, where phase accuracy governs the imaging condition, retains full precision.

The four criteria deliberately cover the two families of indicators identified in the AMR literature [28]: the boundary, energy, and wavefront-trailing criteria rest on local features of the field and are cheap to evaluate, whereas the sensitivity criterion is adjoint-guided, more expensive but targeted at the quantity of interest. Indicators grounded in physical structure, rather than in raw gradients, have already demonstrated cost reductions of up to 70% in AMR for fluid flows [13], which supports the methodological bet of this phase.

For each criterion, we implement compensated (Kahan) summation in the low-precision tiles' time-stepping update to control round-off accumulation over the full simulation, following the mechanism identified by Gao and Harms [12]. We explicitly test for a "precision-interface" artifact, analogous to coarse-fine interface reflections in mesh-based AMR, via residual analysis at the boundaries between tiles of differing precisions.

To isolate the effects of mixed precision from the effects of lossy compression before unifying them into a single per-tile error budget, each candidate tagging region is evaluated under four conditions, always compared against the FP64 reference by PSNR, SSIM, and mean relative error, on both the raw wavefield and the final migrated image:

1. full precision, no compression (control);
2. full precision, with differential compression (isolates the compression effect);
3. reduced precision, no compression (isolates the precision effect);
4. reduced precision, with differential compression (the condition actually deployed in production).

Comparing conditions (ii)+(iii) against (iv) indicates whether the two effects are approximately additive, whether one masks the other, or whether there is amplification through interaction. Only then does each tile jointly receive a floating-point format and a checkpoint compression policy (full snapshot versus compressed delta), as a single, jointly tuned error budget. This phase subsumes differential checkpointing, which becomes the storage mechanism for P-AMR-tagged tiles rather than a separate deliverable.

3.4 Phase 4 — Precision-AMR without Native FP64 (M19–M27)

Challenge: GPUs geared primarily toward AI, such as the RTX 5090, offer negligible native FP64 throughput. Emulating FP64 across the entire domain via the Ozaki scheme [25] is technically viable but expensive enough to make indiscriminate use impractical, which only reinforces the need to apply it selectively.

Method: the tagging criteria validated in Phase 3 are applied on the RTX 5090. Per tile, the pipeline selects between native low-precision tensor-core execution, cuBLAS mixed-precision GEMM (FP32 compute with FP64 accumulation) or cuSOLVER iterative refinement for dense sub-kernels, and Ozaki-scheme FP64 emulation for tiles flagged as accuracy-critical.

- This directly operationalizes the motivation for exploiting AI-first GPUs that lack native FP64: expensive emulation is concentrated only where P-AMR's criteria indicate it is needed.
- We quantify the speedup/accuracy trade-off relative to a uniform-Ozaki baseline (emulating FP64 everywhere) to determine how much selective tiling actually saves.
- All errors are bounded jointly (compression $\times$ precision interaction, following the Phase 3 methodology) against the FP64 baseline using PSNR, SSIM, and mean relative error, across the same six datasets and three checkpointing algorithms (Revolve, zCut, Uniform) used throughout the project.

3.5 Phase 5 — Cross-Architecture Validation (AMD Instinct MI300A) (M26–M31)

Challenge: the tagging criteria are motivated by the physics of wave propagation and adjoint sensitivity, not by any particular vendor's arithmetic units. That portability, however, is a claim that needs to be tested, not merely assumed. The concrete obstacle is the compression stack: GPUZIP today compresses checkpoints with Bitcomp, which is NVIDIA-proprietary and has no HIP port, so there is no checkpoint compression support on AMD GPUs. Generic *lossless* compression (LZ4, Snappy, Cascaded) does not solve the problem: Künas et al. [14] ported these algorithms from CUDA to HIP and measured, on AMD CDNA and RDNA GPUs (including the MI300X, of the same CDNA3 family as the MI300A), a compression ratio of only 1.03 to 1.04$\times$ for floating-point seismic data, regardless of architecture. This is why GPUZIP uses bounded-error lossy compression, and it is this component that must be rebuilt for AMD. The difficulty is not specific to this project either: retaining performance portability in adaptive codes in the face of increasing hardware heterogeneity is recognized as an open problem by the AMR community itself [10].

Method: Phase 5 begins by incorporating into GPUZIP a compression library with native AMD support, replacing the Bitcomp/nvCOMP path on the CDNA3 architecture. The options under consideration are:

- **hipCOMP-core** [2]: a HIP port of the same algorithms as nvCOMP (LZ4, Snappy, Cascaded), with HIP/AMD and HIP/CUDA backends. It is the direct analogue of nvCOMP, but still *early-access* and not optimized for AMD GPUs, and it covers only the *lossless* case.
- **HPDR** [6]: a portable framework with native adapters for OpenMP, CUDA, and HIP, implementing pipelines of MGARD (multilevel *lossy* compression with error bounded in quantities of interest), ZFP (fixed-rate compression for multidimensional arrays), and Huffman. It was evaluated on up to 1,024 nodes of Frontier (AMD Instinct MI250X GPUs), reaching data reduction throughput of up to 103 TB/s. It is the most mature functional equivalent of Bitcomp on Instinct hardware, covering the bounded-error *lossy* compression role.

The final choice between these alternatives, or another that emerges, is made at the start of this phase, according to the maturity of AMD compression libraries at that time. The current guidance is to target HPDR with an MGARD or ZFP pipeline as the Bitcomp replacement. With the compression stack ported, the validation proper proceeds:

- We port the same tagging and tiling logic to Santos Dumont's AMD Instinct MI300A nodes via ROCm/HIP (rocBLAS, hipBLAS, and AMD's own reduced-precision matrix paths), without assuming feature parity with the CUDA-specific path.
- We test whether the Phase 3 criteria and the precision $\times$ compression factorial methodology hold on a CDNA3 architecture, re-running the same evaluation protocol used on the NVIDIA GPUs, now with the AMD compressor in place of Bitcomp.
- We test whether the *a priori* estimator built in Phase 2, derived from the velocity model and architecture-independent, transfers directly, without retuning, to the MI300A.
- We compare the speedup/accuracy trade-off achieved on the MI300A against the NVIDIA baselines, to characterize how much of the gain is architecture-independent and how much is specific to the CUDA/Ozaki/nvCOMP stack.

3.6 Phase 6 — Spectral Operators, Integration, and Release (M31–M36)

Challenge: consolidating the supporting techniques (differential checkpointing, mixed-precision GEMM, Ozaki-scheme emulation, spectral operators) and the Precision-AMR module into a single, coherent, reusable software artifact, the P-AMR library described in Expected Results, rather than leaving them as isolated prototypes from each phase.

Method:

- Benchmark cuFFT-based pseudo-spectral derivative operators as an alternative update rule for selected, precision-tagged tiles, as a secondary technique complementing the core P-AMR contribution.
- Integrate the Precision-AMR module, already as an independent library, differential checkpointing, and mixed-precision paths into a unified GPUZIP v3.0 release.
- Multi-node scaling evaluation on Santos Dumont ($\leq$8 GPUs).
- Final analysis, write-up, and submission, with Precision-AMR as the central scientific claim and differential checkpointing, mixed-precision linear algebra, and spectral operators presented as supporting techniques.

3.7 Computational Platforms

HPC SDK (nvc++/nvfortran, CUDA-X math libraries, Nsight Systems/Compute). Experience: team uses CUDA and Nsight Systems routinely. Adoption: migrate the Awave-3D build from clang to nvc++ and validate numerical parity against existing baselines. Nsight Compute provides kernel-level roofline analysis, now applied per spatial tile, to identify which tiles are memory-bandwidth-bound versus compute-bound and to objectively scope where a lower-precision tile actually pays off.

RTX 5090 (dedicated hardware, requested, 32 GB GDDR7 plus a server with 64 GB of RAM). This GPU is also based on the Blackwell architecture and shares with same-generation professional cards the same memory bandwidth (1,792 GB/s) and the same 5th-generation tensor cores, with FP8/FP4 support and negligible native FP64 throughput. For a memory-bandwidth-bound finite-difference solver like ours, it is memory bandwidth, not raw CUDA-core count, that determines performance. This makes the RTX 5090 a representative development platform for P-AMR, at a fraction of the cost of an equivalent professional card and within the per-item equipment ceiling set by FAPESP's Regular Research Grant rules for this type of request. The absence of fast native FP64 is not a new disadvantage: it is exactly the precision regime that P-AMR already assumes and exploits, via Ozaki-scheme emulation and mixed-precision GEMM, so the scientific question this project answers, precisely which tiles need emulated FP64 and which do not, remains the same. Its 32 GB of GDDR7 (without ECC) support kernel iteration and the tuning of tagging criteria on representative subsets of the domain; as dedicated, queue-free hardware, it enables fast iterative debugging that a shared batch system is not well suited for. FP64 ground truth and production validation continue to be generated on Santos Dumont, with ECC-protected memory.

Santos Dumont (LNCC/SINAPAD, Tier-0 national supercomputer, Petrópolis-RJ). Provides the large-scale validation environment. Its BullSequana X1000 partition includes 94 nodes with 4× NVIDIA Volta V100 GPUs each, matching the hardware used for the original GPUZIP baselines, plus a dedicated AI node with 8× V100 16GB over NVLink. Its newer BullSequana XH3000 partition adds 62 nodes with 4× NVIDIA Hopper H100 SXM 80GB over NVLink, 36 NVIDIA Grace-Hopper GH200 superchip nodes, and 18 nodes with 2× AMD Instinct MI300A APUs (CDNA3 GPU cores, 128 GB HBM3 each). This heterogeneity lets us generate FP64 ground truth and run the full P-AMR criteria sweep at scale on NVIDIA hardware, and separately test whether the tagging methodology and block-tiling approach generalize to a non-NVIDIA architecture.

nvCOMP / Bitcomp (GPU-side lossy compressor). Experience: integrated and validated in GPUZIP v2.0, where Bitcomp delivered the best accuracy-compression balance on V100. Adoption here: the storage mechanism for differential checkpointing, now indexed by the same tile identifiers used for precision tagging, so each tile carries a single, jointly-tuned precision-and-compression error budget.

cuBLAS (mixed-precision extensions) and cuSOLVER (iterative refinement). Both ship inside the HPC SDK. Experience: team has used BLAS/LAPACK routines in prior HPC work. Adoption: applied only within tiles that P-AMR flags as containing dense subkernels needing accuracy beyond native low precision (FP32 compute with FP64 accumulation, and cuSOLVER iterative refinement).

Ozaki-scheme FP64 emulation via low-precision tensor cores. The scheme, introduced by Ozaki et al. [19], decomposes a matrix product into a sum of error-free products, which makes it possible to recover high-precision accuracy from lower-precision routines. Adoption: reserved for the subset of tiles that P-AMR's sensitivity criteria mark as accuracy-critical (near the wavefront, high adjoint-sensitivity magnitude), rather than applied uniformly. This directly tests whether selective emulation is cheaper than emulating everywhere while preserving image fidelity. Recent work has demonstrated up to 2.3× speedup over native FP64 DGEMM on Blackwell GPUs [25], measured on the professional RTX PRO 6000 card; we expect comparable results on the RTX 5090, since both share the same tensor-core generation.

cuFFT. Experience: used in prior spectral operator work. Adoption: benchmarked as an alternative to FDM stencils for wavefield updates within selected, precision-tagged tiles, characterising the memory-bandwidth versus compute trade-off relative to the standard finite-difference path.

AMD Instinct MI300A (CDNA3, Santos Dumont, exploratory). As an additional validation activity, we assess how much of the P-AMR framework, that is, the block/tile decomposition, the four tagging criteria, and the joint precision-compression error budget, transfers to AMD's ROCm/HIP ecosystem (rocBLAS, hipBLAS, and AMD's own reduced-precision matrix paths), without assuming feature parity with the CUDA-specific Ozaki-scheme emulation or nvCOMP/Bitcomp compression path. This last gap is concrete: there is no HIP port of Bitcomp, hipCOMP-core [2] is still *early-access*, and generic *lossless* compression yields only 1.03 to 1.04$\times$ on floating-point seismic data [14], which motivates using a portable bounded-error *lossy* compressor such as HPDR [6] in Phase 5. The goal is to determine whether Precision-AMR is a portable methodology or an NVIDIA-specific optimization.

4 Execution Timeline

The project runs for 36 months, organized in the six phases described in the previous section. Completion is planned for month 36, with paper submission and the GPUZIP v3.0 release. Each phase ends with a verifiable deliverable that serves as a progress milestone; the v3.0-alpha release at the end of Phase 3 (month 20) is the main intermediate milestone.

Phase	Months	Main activities	Deliverable at the end of the phase
1	M1–M4	Port Awave-3D and GPUZIP to the HPC SDK (nvc++); re-integrate nvCOMP/Bitcomp; reproduce the V100 baselines on Santos Dumont; establish the block/*tile* decomposition with a per-block precision tag; per-block *roofline* profiling (Nsight Compute) on the RTX 5090 and Santos Dumont.	GPUZIP/Awave-3D ported and validated under the HPC SDK, with numerical parity confirmed against the V100 baselines and the block decomposition extended to carry precision *tags*; per-block *roofline* profiling report.

Phase	Months	Main activities	Deliverable at the end of the phase
2	M4–M10	Instrument the solver to record FP dynamic-range statistics (exponent histogram, minimum/maximum magnitude, variance) per block and per time step from FP64 reference runs on the six datasets; correlate with velocity-model attributes; build the *a priori* precision-zoning estimator; validate or refute the physical assumptions of the four criteria.	FP dynamic-range characterization database (space-time, six datasets) and *a priori* precision-zoning estimator derived from the velocity model, with a report on which model attributes are predictive and which of the four criteria's assumptions hold empirically.
3	M10–M20	Design, implement, and validate the four tagging criteria (boundary/PML, energy, adjoint sensitivity, wavefront-trailing) informed by Phase 2; compensated (Kahan) summation in the low-precision tiles; precision-interface artifact testing; precision $\times$ compression factorial study (four conditions) to define the unified per-tile error budget.	Precision-AMR module with the four criteria implemented and validated (PSNR, SSIM, and mean relative error vs. FP64 on the wavefield and the migrated image), documented precision $\times$ compression factorial methodology, and differential checkpointing integrated as the storage mechanism for tagged tiles; open-source GPUZIP v3.0-alpha release and reproducibility package (REDU).

Phase	Months	Main activities	Deliverable at the end of the phase
4	M19–M27	Apply the validated criteria on the RTX 5090; per-tile *pipeline* selecting between native tensor cores, mixed-precision GEMM (cuBLAS), iterative refinement (cuSOLVER), and Ozaki-scheme FP64 emulation in accuracy-critical tiles; quantify the performance $\times$ accuracy trade-off against the uniform-Ozaki baseline; bound errors jointly (compression $\times$ precision) across the six datasets and three checkpointing algorithms.	Operational Precision-AMR *pipeline* on hardware without native FP64, with the selective-emulation gain quantified against the uniform-Ozaki baseline and errors bounded vs. FP64 across the six datasets.
5	M26–M31	Incorporate into GPUZIP a compression library with native AMD support (choice between hipCOMP-core, HPDR with MGARD/ZFP, or another, decided at the start of the phase); port the tagging and *tiling* logic to the AMD Instinct MI300A nodes via ROCm/HIP; re-run the evaluation protocol (criteria and precision $\times$ compression factorial) with the AMD compressor; test direct transfer of the Phase 2 *a priori* estimator; compare performance $\times$ accuracy against the NVIDIA baselines.	GPUZIP with a working AMD compression path (CDNA3) and a Precision-AMR portability report for the MI300A, characterizing how much of the gain is architecture-independent and how much is specific to the CUDA/Ozaki/nvCOMP stack.

6	M31–M36	Benchmark cuFFT-based pseudo-spectral operators as an alternative update rule in tagged tiles; integrate the Precision-AMR module (as an independent library), differential checkpointing, and mixed-precision paths into a unified GPUZIP v3.0 release; multi-node scaling evaluation ($\leq$8 GPUs) on Santos Dumont; final analysis, write-up, and submission.	GPUZIP v3.0 release (MIT license, GitHub) with the Precision-AMR library, the cuFFT operator *benchmark*, and the multi-node scaling evaluation; submitted paper and final public reproducibility package (REDU).

5 Description of the Activities Carried Out by the Team

[Placeholder — section to be written last. It will describe, in up to one paragraph per person, the activities of the Principal Investigator, the Associated Researchers, the requested scholarship holders, and the other collaborators not funded by FAPESP.]

6 Evaluation and Dissemination

This section describes how the project's results will be evaluated and disseminated. Evaluation covers the datasets and the experimental protocol; dissemination covers the channels for scientific publication, open-source software, and outreach.

6.1 Datasets

The experiments use six datasets. Five are open synthetic models, widely used as benchmarks by the seismic imaging community, and the sixth is a large in-house model, used mainly in the scalability tests.

The term *SEG/EAGE open data* [22] denotes the body of synthetic velocity models and seismic datasets released publicly by the *Society of Exploration Geophysicists* (SEG) and the *European Association of Geoscientists and Engineers* (EAGE) for benchmarking processing and imaging algorithms (migration, velocity analysis, multiple suppression), for the study of wave propagation in complex geological media, and for comparing hardware platforms. These data are non-confidential and free for research use.

- **Marmousi3D:** a three-dimensional extension of the Marmousi model [27], a synthetic model of complex geological structure, with intense faulting and strong lateral velocity variation, which has become a standard benchmark for FWI and RTM.

- **SEG/EAGE Salt:** a 3D model with a large, complex-geometry salt body over a faulted sedimentary sequence, part of the SEG/EAGE 3-D Modeling Series [3]. It is the canonical subsalt imaging benchmark.
- **SEG/EAGE Overthrust:** a 3D model of a thrust belt (compressional tectonics), with steeply dipping structure and strong lateral velocity variation, also from the SEG/EAGE 3-D Modeling Series [3]. It tests imaging under strong shallow heterogeneity.
- **Sigsbee 2A and Sigsbee 2B:** 2D synthetic subsalt models, inspired by the Sigsbee Escarpment in the Gulf of Mexico and released by the SMAART JV consortium [20]. The 2A model is a constant-density acoustic model without free-surface multiples; the 2B model adds strong internal and free-surface multiples. They are classic benchmarks for subsalt imaging under low signal-to-noise conditions.
- **Large:** a large in-house model (velocities and traces on numerical grids), used mainly in the multi-node scalability tests, as it requires the largest volume of wavefield state and checkpoints.

All are numerical grids of seismic velocity models and traces, with no personal data and no confidentiality restriction.

6.2 Evaluation protocol

All experiments span the six datasets above, three checkpointing algorithms (Revolve, zCut, Uniform), and the four P-AMR tagging criteria (individually and in combination). Kernel-level profiling uses Nsight Compute (roofline, memory-traffic breakdowns); timeline profiling uses Nsight Systems. Implementation is in C/C++ with CUDA under the HPC SDK, with OpenMP Cluster for multi-node scaling. Accuracy is evaluated on both the wavefield and the final RTM image, since a tiling that preserves wavefield PSNR/SSIM may still, in principle, shift the migrated image if error is spatially correlated with reflectors. For the cross-architecture activity, the same accuracy metrics are computed on Santos Dumont's AMD MI300A partition and compared against the NVIDIA baselines, to characterise how much of the achieved speedup and accuracy trade-off is architecture-independent.

To quantify the benefit of adaptivity independently of architecture and implementation, we adopt metrics in the spirit of those proposed by Beisiegel et al. [4] for AMR, correlating numerical error, effective resolution, and run time per degree of freedom. This makes it possible to separate the gain attributable to precision zoning itself from the gain attributable to the implementation on a specific GPU.

6.3 Dissemination strategy

The project's results will be disseminated through four complementary channels:

- **Scientific publications:** papers in relevant conferences and journals in high-performance computing and seismic imaging, with Precision-AMR as the central contribution. The venues under consideration are detailed in the Expected Results section.
- **Open-source software:** the Precision-AMR library and GPUZIP v3.0 will be released into the public domain on GitHub, or an equivalent platform, under the MIT license, together with the reproducibility package (datasets, Nsight profiles, quality metrics) made available via UNICAMP's REDU.
- **Community uptake:** GPUZIP has already been released as open source and has drawn interest from research groups at UFRN, UFRGS, and USP in incorporating GPU-side compression into their own scientific applications. Precision-AMR, delivered as an independent library, broadens this adoption potential beyond RTM/FWI, and tracking these integrations is itself an indicator of the research's dissemination.
- **Outreach:** the Laboratory of Computer Systems (LSC), to which the Principal Investigator belongs, regularly publicizes research results on LinkedIn; this channel will be used to broaden the reach of the publications and of the software release.

7 Project Support Details

We request a dedicated local server equipped with an NVIDIA RTX 5090 GPU (32 GB GDDR7) and 64 GB of RAM, with an estimated cost of up to R$ 55,000 in Brazil, within the per-item equipment ceiling set by FAPESP's Regular Research Grant rules. This GPU shares with same-generation professional Blackwell cards the same memory bandwidth (1,792 GB/s) and the same 5th-generation tensor cores, which makes it representative for kernel development on a memory-bandwidth-bound solver like ours, at a fraction of the cost of an equivalent professional card. Its negligible native FP64 throughput makes it the scientifically relevant target for the selective-emulation study central to Phase 4: it forces the question of whether Precision-AMR's selective use of Ozaki-scheme emulation can match full-domain emulation at a fraction of the cost. As dedicated, queue-free hardware, it also supports the iterative day-to-day development that a shared batch system is not well suited for. Large-scale validation, that is, the P-AMR criteria sweep across datasets and checkpointing algorithms, FP64 baseline generation, multi-GPU scaling, and the AMD MI300A portability test, runs separately on the Santos Dumont supercomputer (LNCC/SINAPAD), accessed through LNCC's own allocation program and outside the scope of this request.

Resource Request

Resource	Request
Physical hardware	1$\times$ server with an NVIDIA RTX 5090 GPU (32 GB GDDR7) and 64 GB of RAM, dedicated local development hardware, used throughout the project (estimated cost: up to R$ 55,000)

Compute time on Santos Dumont is requested separately through LNCC's own project allocation program (SINAPAD) and is not part of this hardware request.

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